

RFQ to BOQ Scope Extraction using NLP system

The *RFQ to BOQ Scope Extraction using NLP* system is an automated solution that **transforms unstructured tender documents into structured Bill of Quantities (BOQ) data**, significantly improving efficiency and accuracy in construction estimation workflows. Using Natural Language Processing (NLP) techniques such as Named Entity Recognition (NER), semantic parsing, and text classification, the system **extracts critical parameters** like **material type, thickness, location, and standards from RFQs and links them into meaningful relationships for BOQ generation**. The workflow includes document ingestion (PDF/OCR), entity and relation extraction, rule-based validation, and structured output generation, reducing manual processing time by up to 70% while minimizing risks such as scope omission, misinterpretation, and contractual disputes. By integrating knowledge bases, ontology models, and advanced AI methods like transformer-based language models, the tool enhances decision-making, consistency, and scalability in tendering, cost estimation, and procurement planning processes.

Academic Papers

1. Zhang & El-Gohary (2015) – Semantic NLP-Based Information Extraction
https://www.researchgate.net/profile/Jiansong-Zhang-2/publication/273024453_Semantic_NLP-Based_Information_Extraction_from_Construction_Regulatory_Documents_for_Automated_Compliance_Checking/links/5831d22108ae138f1c07a010/Semantic-NLP-Based-Information-Extraction-from-Construction-Regulatory-Documents-for-Automated-Compliance-Checking.pdf
2. Sousa et al. (2024) – NLP in Construction Budgeting
<https://www.emerald.com/ci/article/24/7/292/1230787/Automation-of-text-document-classification-in-the>
3. Nabavi et al. (2023) – NLP for BIM Information Extraction
https://www.itcon.org/papers/2023_13-ITcon-Nabavi.pdf

Technical Sources

4. Helium42 (2026) – AI for BOQ Automation
<https://helium42.com/blog/ai-for-construction-boq>
5. DesignDrafter (2026) – Quantity Extraction Automation
<https://designdrafter.com/quantity-extraction-in-construction-how-ai-is-replacing-manual-boq-with-smart-automation-in-2026/>

NLP Foundations

6. GeeksforGeeks – Named Entity Recognition (NER)

<https://www.geeksforgeeks.org/nlp/named-entity-recognition/>

7. Microsoft NLP Recipes – NER using BERT

https://microsoft.github.io/nlp-recipes/examples/named_entity_recognition/

Industry Studies & Insights

8. Yan et al. (2022) – Text Mining in Construction

https://www.researchgate.net/publication/366045971_Overview_and_analysis_of_the_text_mining_applications_in_the_construction_industry/fulltext/63982e4f095a6a7774272942/Overview-and-analysis-of-the-text-mining-applications-in-the-construction-industry.pdf

9. AEC Contracts – Scope Extraction Risks

<https://aecontracts.com/tender-document-ai-scope-extraction/>

Surveys & Reviews

10. Zheng et al. (2023) – Document-Level Information Extraction Survey

<https://arxiv.org/abs/2309.13249>

Videos

1. AI Document Processing (RFQ → BOQ Core Concept)

- ▶ Explains **OCR + NLP + document automation pipeline**
 - ▶ Covers how unstructured documents (RFQs, contracts) are converted into structured data [\[youtube.com\]](https://www.youtube.com/watch?v=1333333333)

2. PDF → Structured Data (RFQ Parsing)

- ▶ End-to-end pipeline: **PDF → OCR → structured JSON/data**
 - ▶ Directly matches RFQ-to-BOQ transformation workflow [\[youtube.com\]](https://www.youtube.com/watch?v=2333333333)
- ▶ AI tool demo converting PDF documents into Excel/structured outputs [\[youtube.com\]](https://www.youtube.com/watch?v=3333333333)

3. NLP Information Extraction (Entity + Relation Extraction)

- ▶ Named Entity Recognition (NER) using BERT
 - ▶ Extracts entities like material, quantity, location — same as BOQ extraction [\[youtube.com\]](https://www.youtube.com/watch?v=4333333333)
- ▶ Practical implementation of **BERT-based entity extraction workflow** [\[youtube.com\]](https://www.youtube.com/watch?v=5333333333)

4. NLP Fundamentals (Understanding Pipeline)

- ► Complete NLP course (tokenization, parsing, classification)
 - Supports document's NLP techniques section [\[youtube.com\]](#)

5. BOQ Automation & Construction AI (Application Layer)

- ► AI-driven **BOQ generation from drawings using natural language** [\[youtube.com\]](#)
- ► Real-world demo: **Manual BOQ → AI automation workflow** [\[youtube.com\]](#)

6. End-to-End AI + BOQ Pipeline (Advanced)

- ► Full pipeline:
BIM/Docs → AI extraction → BOQ → risk/analysis
 - Closely aligns with future system architecture in document [\[youtube.com\]](#)